

Multiplicative Inverse

Module I

Multiplicative Inverse

- A multiplicative Inverse is reciprocal.
- What you multiply by a number to get 1.
- $8 * ? = 1$, Ans: $1/8$
- $\frac{3}{4} * ? = 1$, Ans: $4/3$

Modular Multiplicative Inverse

- A modular multiplicative inverse of an integer a is an integer x such that the product $a * x$ is congruent to 1 with respect to the modulus m .
- $a * x \equiv 1 \pmod{m}$, this can be re written as $a * x \bmod m = 1 \pmod{m}$ or else, $(a * x) \bmod m = 1$.
- Example: $3 * x \equiv 1 \pmod{17} \rightarrow (3 * x) \bmod 17 = 1$, here $x=6$.
- By trail and error method,
- $(3 * 1) \bmod 17 = 3$ $\times$
- $(3 * 2) \bmod 17 = 6$ $\times$
- $(3 * 3) \bmod 17 = 9$ $\times$
- $(3 * 4) \bmod 17 = 12$ $\times$
- $(3 * 5) \bmod 17 = 15$ $\times$
- $(3 * 6) \bmod 17 = 1$

Finding modular multiplicative inverse using extended Euclidean algorithm

- $5x \equiv 1 \pmod{14}$ find x ?

$\gcd(5, 14)$ $\frac{b}{a} \quad b = aq + r$

$5 \times 3 \pmod{14} = 1$
 $15 \pmod{14} = 1$
 $1 = 1$

$$14 = 2 \times 5 + 4$$

$$5 = 1 \times 4 + 1 \checkmark$$

$$4 = 14 - 2 \times 5$$

$$1 = 5 - 1 \times 4$$

$$1 = 5 - 1 \times (14 - 2 \times 5)$$

$$1 = 3 \times 5 + (-1) \times 14$$

$$1 = \underbrace{(3)}_{\frac{1}{a}} \underbrace{5}_{a} + \underbrace{(-1)}_{\frac{y}{b}} \underbrace{14}_{b}$$

$5 * x \equiv 1 \pmod{14}, \text{ find } x$

$\gcd(5, 14) = 1$

Another trick

$35 * x \equiv 1 \pmod{8}$, find x

$$35x \pmod{8} = 1 \pmod{8}$$

$$\frac{35}{8} = 3$$

$$3x \pmod{8} = 1 \pmod{8}$$

$$3 \times 3 \pmod{8} = 1$$

$$9 \pmod{8} = 1$$

$$35 \times \underline{3} \pmod{8} = 1$$

$$105 \pmod{8} = 1$$

$$1 = 1$$

Another trick exercises

40* $x \equiv 1 \pmod{7}$, find x ? *Ans* = 3

20* $x \equiv 1 \pmod{3}$, find x ? *Ans* = 2